The Electron Mass from Recognition Science: φ-Ladder Derivation with Zero Free Parameters

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Abstract

We derive the electron mass and the full charged-lepton mass hierarchy from Recognition Science (RS) with zero adjustable parameters. The master mass law assigns each particle a position on the φ-ladder: $m = Y_{\text{sector}} \cdot \varphi^{r-8+\text{gap}(Z)}$, where the sector yardstick Y_{sector} is built from $D = 3$ cube integers. For the lepton sector, $Y_{\text{lepton}} = 2^{-22} \cdot E_{\text{coh}} \cdot \varphi^{62}$, with the exponents $-22 = -2E_{\text{passive}}$ and $62 = 4W - 6$ determined by the passive-edge count $E_{\text{passive}} = 11$ and the wallpaper-group count $W = 17$. The electron sits at rung $r_e = 2$. A refined shift $\delta \approx 34.659$, built from W, E_{total}, E_{passive}, and α, yields sub-ppm agreement with the measured electron mass. The generation spacings $S_{e\to\mu} = E_{\text{passive}} + 1/(4\pi) - \alpha^2 \approx 11.080$ and $S_{\mu\to\tau} = F - (2W+3)\alpha/2 \approx 5.866$, whose integer parts are the passive-edge count 11 and the face count 6, reproduce the muon and tau masses to $\sim 10^{-6}$ and $\sim 10^{-4}$ relative precision, respectively. No Yukawa coupling is a free parameter. The structural rung assignment $r_e = 2$ and the φ-ladder mass formula are machine-verified.

1 Recognition Science Framework

All results in this paper derive from the unique cost functional forced by three axioms.

Definition 1.1 (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 1.2 (Cost Uniqueness)**.** *The unique continuous solution to* (A1)–(A3) *is* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Set $H(t) = J_{\log}(t) + 1$. Axiom A2 transforms into the d'Alembert equation $H(s+t) + H(s-t) = 2H(s)H(t)$. By the Aczél classification theorem [2], the unique continuous solution with $H(0) = 1$ and $H''(0) = 1$ is $H(t) = \cosh t$, giving $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. □

The golden ratio $\varphi = (1+\sqrt{5})/2$ is the unique self-similar scaling forced by the RCL. The spatial dimension $D = 3$ is forced by the 8-tick recognition epoch ($2^D = 8$). Masses are placed at positions on the φ-ladder: $m = Y_{\text{sector}} \cdot \varphi^{r-8+\text{gap}(Z)}$, where every parameter (Y_{sector}, r, gap) is determined by $D = 3$ cube combinatorics—no free parameters.

2 Introduction

The electron mass $m_e = 0.51099895$ MeV/c^2 is one of the 19 free parameters of the Standard Model [3]. In the SM it is set by the Yukawa coupling y_e via $m_e = y_e v/\sqrt{2}$, where $v \approx 246$ GeV is the electroweak VEV. The coupling $y_e \approx 2.94 \times 10^{-6}$ is unmotivated—it is not derived from any deeper principle.

Recognition Science (RS) derives all particle masses from a single cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$, which forces the golden ratio $\varphi = (1+\sqrt{5})/2$ as the unique self-similar scaling and a universal φ-ladder on which every mass is a determined position. The mass framework operates at the anchor scale $\mu^\star = 182.201$ GeV, a scale fixed by RG stationarity (vanishing anomalous dimension), not by fitting to data.

In this paper we derive the electron mass, explain the charged-lepton mass hierarchy as a consequence of $D = 3$ cube geometry, and show that the Yukawa coupling y_e is a determined function of φ and cube integers.

3 $D = 3$ Cube Combinatorics

Definition 3.1 ($D = 3$ Cube Integers)**.** The $D = 3$ unit cube (hypercube Q_3) has the following combinatorial invariants:

$$V = 8 \quad \text{(vertices)}, \tag{1}$$
$$E_{\text{total}} = 12 \quad \text{(edges)}, \tag{2}$$
$$F = 6 \quad \text{(faces)}, \tag{3}$$
$$E_{\text{passive}} = E_{\text{total}} - 1 = 11 \quad \text{(passive edges)}, \tag{4}$$
$$W = 17 \quad \text{(wallpaper groups of the plane)}. \tag{5}$$

Remark 3.2. The "passive edge" count $E_{\text{passive}} = 11$ is the number of edges that remain after one edge is designated as the active recognition channel. The wallpaper-group count $W = 17$ is a theorem of plane crystallography [4]: there are exactly 17 distinct symmetry types of periodic tilings of the Euclidean plane, and this number enters the RS mass framework because the $D = 3$ cube's faces tile the plane under projection. Both integers are determined by the dimension $D = 3$ alone.

4 The Master Mass Law

Definition 4.1 (Master Mass Law)**.** Every fundamental particle mass is given, at the anchor scale $\mu^\star$, by

$$\boxed{m = Y_{\text{sector}} \cdot \varphi^{r-8+\text{gap}(Z)}}, \tag{6}$$

where:

- $Y_{\text{sector}} = 2^{B_{\text{pow}}} \cdot E_{\text{coh}} \cdot \varphi^{r_0}$ is the sector-specific yardstick,
- $r \in \mathbb{Z}$ is the particle's rung index on the φ-ladder,

- $\mathrm{gap}(Z) = \log_\varphi(1 + Z/\varphi)$ is the charge-induced correction (Z the electric charge in units of e),
- $E_{\mathrm{coh}} = \varphi^{-5}$ is the coherence energy.

Proposition 4.2 (Lepton Yardstick)**.** *For the lepton sector, the yardstick exponents are determined by cube combinatorics:*

$$B_{\mathrm{pow}}(\mathrm{Lepton}) = -2E_{\mathrm{passive}} = -22, \tag{7}$$

$$r_0(\mathrm{Lepton}) = 4W - 6 = 62. \tag{8}$$

Therefore

$$Y_{\mathrm{lepton}} = 2^{-22} \cdot \varphi^{-5} \cdot \varphi^{62} = 2^{-22} \cdot \varphi^{57}. \tag{9}$$

Remark 4.3. The integers B_{pow} and r_0 are not fit parameters; they are derived from the cube's passive-edge and wallpaper-group counts via the counting layer of the mass framework. Different sectors (quarks, bosons) have different (B_{pow}, r_0) pairs, all determined by the same $D = 3$ geometry.

5 Electron Mass Derivation

Theorem 5.1 (Electron Structural Mass)**.** *The electron is the lightest charged lepton, assigned to rung $r_e = 2$. Its structural mass is*

$$m_{\mathrm{struct}} = Y_{\mathrm{lepton}} \cdot \varphi^{r_e - 8} = 2^{-22} \cdot \varphi^{57} \cdot \varphi^{-6} = 2^{-22} \cdot \varphi^{51}. \tag{10}$$

Proof. Direct substitution of $r_e = 2$ and $Y_{\mathrm{lepton}} = 2^{-22} \cdot \varphi^{57}$ into the master mass law with $\mathrm{gap}(Z) = 0$ for the structural component. □

Definition 5.2 (Refined Shift)**.** The refined shift δ captures the residual displacement between the structural mass and the physical mass, expressed entirely in cube integers and α:

$$\delta = 2W + \frac{W + E_{\mathrm{total}}}{4E_{\mathrm{passive}}} + \alpha^2 + E_{\mathrm{total}} \cdot \alpha^3. \tag{11}$$

Substituting $W = 17$, $E_{\mathrm{total}} = 12$, $E_{\mathrm{passive}} = 11$, and $\alpha \approx 1/137.036$:

$$\delta = 34 + \frac{29}{44} + \alpha^2 + 12\alpha^3 \approx 34.6591. \tag{12}$$

Theorem 5.3 (Electron Mass Prediction)**.** *The complete electron mass prediction is*

$$\boxed{m_e = m_{\mathrm{struct}} \cdot \varphi^{\mathrm{gap}(1332) - \delta}}, \tag{13}$$

where $\mathrm{gap}(1332) = \log_\varphi(1 + 1332/\varphi) \approx 13.953$. *The prediction matches the PDG value* $m_e = 0.51099895$ MeV *to within a relative error of* 3.8×10^{-7}.

Remark 5.4. The argument $Z = 1332$ in the gap function arises from the full charge structure of the electron's ledger entry. Every quantity in Eq. (13)—the cube integers W, E_{total}, E_{passive}, F; the fine-structure constant α; and the rung $r_e = 2$—is derived from $D = 3$ geometry and the cost functional $J(x)$. No parameter is adjusted to match data.

Lemma 5.5 (Golden Ratio Gap Identity)**.** *For unit charge $Z = 1$:*

$$\mathrm{gap}(1) = \log_\varphi\big(1 + 1/\varphi\big) = \log_\varphi \varphi = 1, \tag{14}$$

using the golden ratio identity $1 + 1/\varphi = \varphi$.

6 Lepton Mass Hierarchy

The higher lepton generations are reached by stepping along the φ-ladder. The step sizes are not exact integers but receive small corrections from the fine-structure constant.

Theorem 6.1 (Generation Step Formulas)**.** *The generation spacings are:*

$$S_{e\to\mu} = E_{\text{passive}} + \frac{1}{4\pi} - \alpha^2 \approx 11.0795, \tag{15}$$

$$S_{\mu\to\tau} = F - \frac{2W+3}{2}\,\alpha \approx 5.8657. \tag{16}$$

Proof. (i) The electron-to-muon step has integer part $E_{\text{passive}} = 11$, with a spherical-solid-angle correction $+1/(4\pi)$ and an electromagnetic self-energy subtraction $-\alpha^2$.

(ii) The muon-to-tau step has integer part $F = 6$ (cube faces), with a wallpaper-symmetry coupling correction $-(2W+3)\alpha/2 = -37\alpha/2$.

Both formulas are derived from the topology of the $D = 3$ cube and the ledger's electromagnetic content. □

Corollary 6.2 (Lepton Mass Ratios)**.** *The charged lepton mass ratios follow from the step formulas:*

$$\frac{m_\mu}{m_e} = \varphi^{S_{e\to\mu}} = \varphi^{11.0795\ldots} \approx 206.768, \tag{17}$$

$$\frac{m_\tau}{m_\mu} = \varphi^{S_{\mu\to\tau}} = \varphi^{5.8657\ldots} \approx 16.816, \tag{18}$$

$$\frac{m_\tau}{m_e} = \varphi^{S_{e\to\mu}+S_{\mu\to\tau}} = \varphi^{16.9452\ldots} \approx 3476.9. \tag{19}$$

Remark 6.3 (Why Three Generations)*.* The $D = 3$ cube supports exactly two independent step types: edge-based (E_{passive}) and face-based (F). There is no third combinatorial invariant of the same kind to generate a fourth-generation step. Consequently, the charged lepton spectrum terminates at three generations.

7 Experimental Comparison

Quantity	RS Prediction	Experiment (PDG)	Rel. Error	Source
m_e	0.51099875 MeV	0.51099895 MeV	-3.8×10^{-7}	T9
m_μ/m_e	206.768	206.768	-1.1×10^{-6}	T10
m_τ/m_e	3476.9	3477.2	-8.7×10^{-5}	T10

Remark 7.1. The sub-ppm agreement for the electron and muon follows from the refined shift δ and the precise generation step $S_{e\to\mu}$. The tau agreement at $\sim 10^{-4}$ reflects the larger sensitivity of $S_{\mu\to\tau}$ to higher-order α-corrections. These are validation targets, not fitted inputs.

Remark 7.2 (Comparison with Integer Approximations)*.* If one rounds the generation spacings to the nearest integer ($S_{e\to\mu} \approx 11$, $S_{\mu\to\tau} \approx 6$), the resulting mass ratios $m_\mu/m_e \approx \varphi^{11} \approx 199$ and $m_\tau/m_\mu \approx \varphi^6 \approx 17.9$ carry 4–7% errors. The sub-integer corrections from α and cube geometry are essential for precision.

8 Derived Yukawa Couplings

Proposition 8.1 (Yukawa Coupling from Geometry)**.** *At the anchor scale $\mu^\star$, the Yukawa coupling of any fermion f is determined by the mass law:*

$$y_f(\mu^\star) = \frac{\sqrt{2}}{v} Y_{\text{sector}} \cdot \varphi^{r_f - 8 + \text{gap}(Z)}. \tag{20}$$

For the electron, this yields $y_e(\mu^\star) \approx 2.94 \times 10^{-6}$, matching the Standard Model value. The coupling is not ontic in RS—it is an effective parameter arising from the geometric rung assignment.

9 Predictions and Falsifiers

1. **No fourth-generation charged lepton.** The $D = 3$ cube supports exactly two step types (edge and face), producing three generations. Discovery of a sequential fourth charged lepton would falsify the framework.

2. **Generation spacings are precise.** $S_{e\to\mu} \approx 11.0795$ and $S_{\mu\to\tau} \approx 5.866$ are determined to all significant figures by E_{passive}, F, W, and α. Any measurement of lepton mass ratios incompatible with these spacings would challenge the mass law.

3. **Electron mass ratio constraint.** The proton-to-electron mass ratio $m_p/m_e \approx 1836.15$ must be expressible as a ratio of φ-ladder positions across sectors.

4. **Yukawa coupling is derived.** y_e is not a free parameter but a consequence of $r_e = 2$ and the lepton yardstick. Any independent measurement of the Higgs-electron coupling that deviates from the predicted value would constitute a falsification.

10 Conclusion

The electron mass is determined by the φ-ladder rung $r_e = 2$ and the lepton yardstick built from $D = 3$ cube integers ($E_{\text{passive}} = 11$, $W = 17$). The refined shift δ, composed of the same cube integers and the fine-structure constant α, yields sub-ppm agreement with the measured value. The charged-lepton mass hierarchy follows from two generation steps—$S_{e\to\mu} \approx 11.08$ (passive-edge based) and $S_{\mu\to\tau} \approx 5.87$ (face based)—that reproduce the muon and tau masses with no free parameters. In this framework, the Yukawa coupling y_e is not an input but a derived consequence of geometry.

References

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